Tiv Source: Product and Technical Specification

Nov 6, 2026 · @Luke

Tiv Source is an Android app that gives Middle Belt communities everyday social messaging, hidden invite-only spaces for verified reporting, and a built-in Tiv-language AI assistant.

1. Overview and scope

Tiv Source gives communities one app for daily social life and, for approved members, private spaces where reports are captured, verified and escalated without exposing who made them.

Goals

- G1: Protect the identity of reporters and verifiers.
- G2: Make shared reports verifiable against AI-generated media, distortion and recycled old content.
- G3: Work as an ordinary social app, so owning it does not mark a person as a target.
- G4: Provide a conversational AI in the Tiv language, built into the app.
- G5: Run on low-cost Android phones and weak connections.

Platform: Android (version 8 and above) is the only target for version 1.

Design principles

1. One app for everyone. Access depends on keys and approvals, not on which build a person installed.
2. Private spaces are an ordinary, documented feature of the app.
3. Provenance over detection. Media captured in the app is signed at capture. Everything else is labeled unverified.
4. Every report is reviewed by a verifier before it is escalated.
5. Reports describe observable events and behavior, never named private individuals or groups.
6. Voice and message data for AI features stays on the device or on project-controlled servers.
7. The code is open source.

2. Users and roles

Five roles control what each account can do. Role data is stored in encrypted room state, never in public profiles.

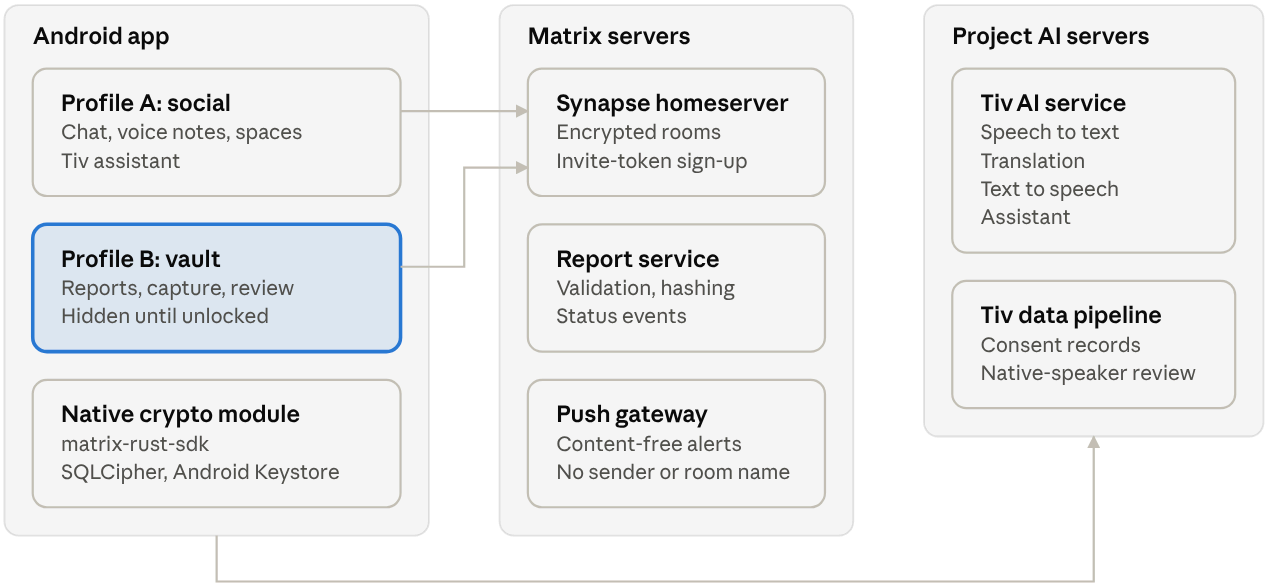
Role	Social spaces	Vault access	Submit reports	Review reports	Approve escalation	Manage invites and roles
Member	Yes	No	No	No	No	No
Reporter	Yes	Yes	Yes	No	No	No
Verifier	Yes	Yes	Yes	Yes	No	No
Desk administrator	Yes	Yes	Yes	Yes	Yes	Yes
Operator	No content access	No	No	No	No	Server administration only

Role rules

- Every new account starts as a Member.
- A Member becomes a Reporter only through approval by a desk administrator or vouching by an existing Reporter, recorded in encrypted state.
- Only a desk administrator can assign the Verifier or Desk administrator role.
- An Operator can run servers but must not be able to read message content, reports or proof bundles.
- Any role can be revoked. Revocation triggers key rotation for every room the account belonged to.

3. System architecture

Two profiles share one phone but no keys, and AI runs only on project servers



AI requests go to project servers only, and the assistant never sees vault content

system architecture · app, Matrix servers, AI servers

The app holds two separate profiles, and both talk to the Matrix servers. Tiv AI requests go to project-controlled servers, and the assistant never reads vault content.

4. Profiles and private spaces

Each device holds two separate profiles, so the vault never shares an account, key or database with the social app.

Item	Profile A (social)	Profile B (vault)
Matrix account	Own account	Separate account
Device keys	Own keys	Separate keys
Local database	Own encrypted database	Separate encrypted database
Loaded at launch	Yes	No, only after unlock
Listed in the interface	Yes	No, until unlocked
Unlock secret	Default PIN or biometric	Separate passphrase or biometric

Unlock and protection behavior

Trigger	Result
Normal PIN	Opens Profile A
Vault passphrase or biometric	Opens Profile B
Duress PIN	Opens Profile A only. Vault data is not decrypted, listed or synced
Quick wipe gesture	Deletes Profile B data and keys. Profile A is untouched
Inactivity timeout	Locks the profile that is open
Notification arrives	Shows generic text only, with no sender, room name or content

The vault database key is derived from the vault passphrase and is never stored in the clear. Vault screens block screenshots and screen recording.

5. Functional requirements

Priority: M = must have for version 1, S = should have.

5.1 Accounts and onboarding

ID	Requirement	Priority
FR-01	Registration requires a single-use invitation token. No phone number or email is collected.	M
FR-02	The user chooses a pseudonymous display name. There is no real-name field.	M
FR-03	Vault enrollment requires approval from a desk administrator or vouching by a Reporter, recorded in encrypted state.	M
FR-04	Vault key backup is off by default. The user may write down a recovery phrase.	M
FR-05	The social profile may offer optional encrypted backup.	S
FR-06	The interface is available in English and Tiv.	M

5.2 Social messaging

ID	Requirement	Priority
FR-10	One-to-one and group chat, end-to-end encrypted by default.	M
FR-11	Voice notes recorded in Opus, encrypted on the device and sent in resumable chunks.	M
FR-12	Image and file sharing with client-side encryption.	M
FR-13	Community spaces with topic rooms (for example Farm, Market, Events, Support).	M
FR-14	Read receipts, typing indicators and last-seen are off by default.	M
FR-15	Offline composing with a send queue that resumes on reconnect.	M
FR-16	Disappearing messages, configurable per room.	S

5.3 Profiles and protection

ID	Requirement	Priority
FR-20	Two separate profiles with separate Matrix accounts, keys and databases (section 4).	M
FR-21	Separate vault passphrase or biometric. Vault data is not decrypted or listed before unlock.	M
FR-22	A duress PIN opens Profile A only.	M
FR-23	A quick wipe gesture deletes vault data and keys.	M
FR-24	Notifications show generic text only.	M
FR-25	Screenshot and screen-recording blocking in the vault.	M
FR-26	Configurable auto-lock after inactivity.	M

5.4 Reporting and capture

ID	Requirement	Priority
FR-30	Structured report form (schema in section 7.1) with offline queueing.	M
FR-31	In-app capture produces a signed media item with a proof bundle: hash, signature, timestamp and device attestation where available.	M
FR-32	Imported or forwarded media is accepted only as unverified, is visibly labeled, and never counts toward corroboration.	M
FR-33	Shared copies of media have EXIF data and precise location stripped. The full proof bundle goes only to verifiers, encrypted.	M
FR-34	Location is stored at ward or LGA level by default. Precise coordinates require explicit reporter consent.	M

ID	Requirement	Priority
FR-35	A reporter can check the status of their own reports through the pseudonymous channel.	M
FR-36	A reporter can add a follow-up note or media to an open report.	S

5.5 Verification and escalation (verifier tools inside the vault)

ID	Requirement	Priority
FR-40	Verifier queue listing reports by status and age.	M
FR-41	Report detail view with proof status, duplicate flags and corroboration count.	M
FR-42	Perceptual hashes of every image and video frame are computed and compared with earlier submissions. Near-duplicates are flagged.	M
FR-43	A verifier can request clarification from the reporter without learning the reporter's identity.	M
FR-44	A verifier can link independent reports of the same event and record corroboration.	M
FR-45	Status changes follow the lifecycle in section 6 and are written to an append-only audit record visible to desk administrators.	M
FR-46	A desk administrator can approve a sanitized escalation summary. The summary contains no reporter identifiers, precise device data or proof bundles.	M
FR-47	Forwarding of unverified reports outside the verification desk is disabled.	M
FR-48	Verifier notes are private to the desk.	M

5.6 Safety guardrails

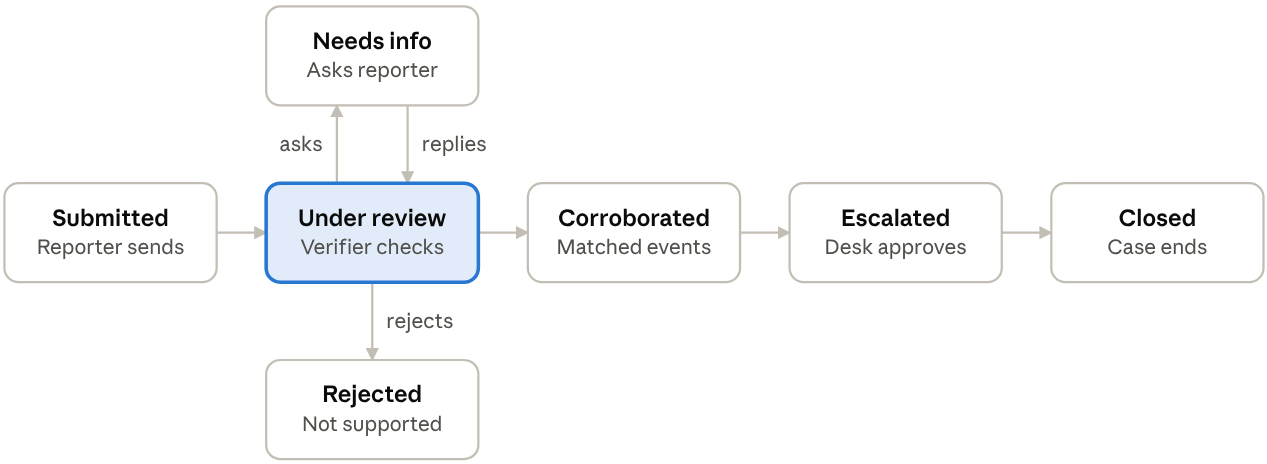
ID	Requirement	Priority
FR-50	The report form has no field for naming private individuals or groups.	M
FR-51	Free text is scanned for likely personal names and group labels, and the reporter is prompted to remove them.	S
FR-52	Abuse reporting and member suspension by desk administrators.	M
FR-53	Rate limits on report submission and invitation creation.	M

5.7 Tiv AI features in the app

ID	Requirement	Priority
FR-60	A Tiv assistant chat available in the social profile.	M
FR-61	Tiv to English and English to Tiv translation of text.	M
FR-62	Speech input in Tiv (speech to text) in the assistant and the message composer.	S
FR-63	Spoken output in Tiv (text to speech) for assistant replies.	S
FR-64	Opt-in transcription and translation of received voice notes, run on the device or on project servers.	S
FR-65	A voice-first mode that lets a user complete core tasks with spoken prompts and audio feedback.	S
FR-66	Opt-in voice contribution module for building the Tiv dataset, with a consent record per speaker (section 8.3).	M
FR-67	The assistant does not read vault content and does not give security advice.	M

6. Report lifecycle

Every report passes verifier review before it can be escalated



report lifecycle · 7 statuses

A verifier reviews every submitted report, can ask the reporter for more information, and moves it to corroborated or rejected. Corroborated reports can be escalated after desk approval, and every status change is written to the audit record.

7. Data models

7.1 Report

No field stores the reporter's identity. The link between a reporter and a report exists only in encrypted room membership and is never exported with an escalation.

```
{
  "report_id": "uuid",
  "created_at": "ISO-8601 timestamp",
  "event_window": { "start": "ISO-8601", "end": "ISO-8601" },
  "category": "movement | attack | kidnapping | extortion | displacement |
other",
  "coarse_location": { "lga_code": "string", "ward": "string" },
  "description": "free text",
  "language": "tiv | en | ha | pcm | other",
  "confidence": "low | medium | high",
  "source_type": "direct_witness | second_hand | media_only",
  "media": [
```

```
    {
      "media_id": "uuid",
      "tier": "captured | imported",
      "sha256": "hex",
      "phash": "hex",
      "proof_bundle_ref": "encrypted blob reference"
    }
  ],
  "status": "submitted | under_review | needs_info | corroborated | rejected
| escalated | closed",
  "corroboration_count": 0,
  "duplicate_of": "report_id or null",
  "linked_reports": ["report_id"]
}
```

7.2 Matrix event types

All structured data uses events in the `org.tivsource` namespace.

Event type	Purpose	Sent by
org.tivsource.report	A submitted report	Reporter
org.tivsource.report.status	A status change with reason	Verifier, desk administrator
org.tivsource.report.note	Verifier note, private to the desk	Verifier
org.tivsource.report.clarify	Clarification request or reply	Verifier, Reporter
org.tivsource.role	Role grant or revocation	Desk administrator
org.tivsource.escalation	Approved sanitized summary	Desk administrator

7.3 Proof bundle

Field	Description
media_sha256	Hash of the original capture
signature	Device-held key signature over the hash and metadata

Field	Description
captured_at	Device timestamp plus trusted timestamp where available
device_attestation	Attestation token from the platform, where available
app_version	Build that produced the capture
location_detail	Precise location, present only with reporter consent

The bundle is encrypted to the verifier room keys. Shared copies of the media never include it.

7.4 Tiv AI records

Record	Fields
ConsentRecord	speaker_id (random), consent_version, purposes, granted_at, revoked_at
SpeechSample	sample_id, speaker_id, prompt_id, dialect, age_band, audio_ref, quality_score, review_status
TextPair	pair_id, tiv_text, en_text, source, reviewer_id, review_status
EvalItem	item_id, task (asr, mt, tts, assistant), input, reference, native_reviewer_scores

8. Tiv AI specification

The Tiv AI is a set of four capabilities served through one interface, built by fine-tuning open models on community-contributed data.

8.1 Capabilities

Capability	Input	Output	Where it runs
Speech to text (ASR)	Tiv audio	Tiv text	Device for short clips, project server for long clips
Translation (MT)	Tiv or English text	English or Tiv text	Project server, with a small on-device model for offline use as a later goal
Text to speech (TTS)	Tiv text	Tiv audio	Project server
Assistant	Tiv or English text or speech	Tiv text and optional audio	Project server

The assistant works through a translation layer. A Tiv message is translated to English, answered by a general language model, and translated back to Tiv. Direct Tiv generation replaces this layer when evaluation shows it meets the targets in 8.4.

8.2 Assistant domains for version 1

- Farming: planting calendars, crop care, storage, market information.
- Health: general guidance and when to go to a clinic.
- Language learning: vocabulary, phrases, reading practice.
- App help: how to use Tiv Source features.

Outside these domains the assistant says it cannot help. It gives no security advice and does not present itself as a source of truth.

8.3 Data collection

Item	Specification
Sources	Volunteer recordings, translated texts, storytelling, community-approved radio and text archives
Consent	Informed, per-speaker, revocable, stored as a ConsentRecord with a random speaker id

Item	Specification
Recording tool	In-app module with scripted prompts, audio quality checks and a review queue
Metadata	Dialect, age band and optional gender. No names, phone numbers or locations tied to a sample
Orthography	Single documented spelling and tone-marking convention, with variants recorded
Review	Native-speaker review of every text pair and a sample of every audio batch
Licensing	Community-governed license so the data benefits the Tiv community
Retention and deletion	A speaker can withdraw consent, and their samples are removed from future training sets

8.4 Quality targets (proposed, to be confirmed with native-speaker baselines)

Capability	Metric	Closed-beta target	Release target
ASR	Word error rate on held-out multi-dialect set	35% or lower	25% or lower
MT	Native-speaker adequacy and fluency, 1 to 5	3.5 or higher	4.0 or higher
TTS	Mean opinion score, 1 to 5	3.5 or higher	4.0 or higher
Assistant	Share of in-domain test prompts answered correctly and safely	85% or higher	92% or higher
Assistant safety	Unsafe or out-of-domain advice in the red-team set	0 critical failures	0 critical failures

8.5 Privacy and handling rules

- AI inference runs on the device or on servers controlled by the project. No voice note or message goes to a third-party AI service.

- The assistant has no access to the vault profile.
- Transcription of received voice notes is opt-in and per conversation. Verifiers may use it on the report media they review, and the transcript is always reviewed by a human before it is relied on.
- Voice recordings are treated as sensitive personal data and are stored encrypted with access limited to the data team.
- Each released model has a model card listing languages, dialects, known limits and evaluation results.

8.6 Build sequence

1. Consent flow, recording tool and text collection live.
2. Baseline evaluation of existing open multilingual speech and translation models on a Tiv test set.
3. Fine-tune translation first, then speech recognition, then text to speech.
4. Connect translation to the assistant and release the narrow domains in 8.2 to the closed pilot.
5. Evaluate direct Tiv generation and replace the translation layer when targets are met.

9. Security and privacy requirements

ID	Requirement
SEC-01	No cryptography is written in-house. Use matrix-rust-sdk, vodozemac and vetted platform APIs.
SEC-02	Local databases are encrypted with SQLCipher. Keys are protected by the Android Keystore.
SEC-03	The vault database key is derived from the vault passphrase with a memory-hard function (Argon2id) and never stored in the clear.
SEC-04	All chats and rooms use end-to-end encryption. Unencrypted rooms cannot be created.
SEC-05	Roles, vouching records and report links live in encrypted room state.

ID	Requirement
SEC-06	Removing a member or revoking a role rotates keys for every affected room.
SEC-07	Push notifications carry no content, sender or room name.
SEC-08	No analytics, advertising or crash-reporting SDKs that send identifiers or content to third parties.
SEC-09	Homeserver logs keep no message content and the shortest practical retention for connection data.
SEC-10	Shared media has EXIF data and precise location stripped before sending.
SEC-11	Captured media is signed on the device with a hardware-backed key where available.
SEC-12	Proof bundles are encrypted to verifier room keys and never included in shared copies.
SEC-13	Escalation summaries are generated from an allow-list of fields and contain no reporter identifiers.
SEC-14	The audit record of status changes is append-only and visible to desk administrators.
SEC-15	Vault screens block screenshots, screen recording and the recent-apps preview.
SEC-16	The quick wipe removes vault data, keys, caches, thumbnails and pending queue items.
SEC-17	The duress PIN path never loads vault data, logs or caches.
SEC-18	Dependencies are pinned and reviewed before they are added. Builds are reproducible where practical, and a software bill of materials is published.
SEC-19	AI services follow the handling rules in section 8.5.
SEC-20	An independent security audit is completed before any deployment beyond the closed pilot.

10. Non-functional requirements

ID	Area	Requirement
NFR-01	Devices	Runs on Android 8 and above with 2 GB of RAM.
NFR-02	Size	APK of 40 MB or less, excluding downloadable AI models.
NFR-03	Startup	Cold start in 4 seconds or less on the reference low-cost device.
NFR-04	Offline	Composing, report drafting and capture work with no connection. Queued items sync when connectivity returns.
NFR-05	Queue	The offline queue holds at least 200 messages or 20 reports, and survives app restarts.
NFR-06	Bandwidth	Images are compressed to about 200 KB or less. Voice notes use a bitrate that keeps one minute under about 300 KB.
NFR-07	Resumable transfer	Interrupted uploads and downloads resume without restarting.
NFR-08	Battery	Background sync uses no more than 5% of battery per day on the reference device.
NFR-09	Language	Full interface in English and Tiv. Text sizes and contrast follow accessibility guidance.
NFR-10	Voice-first	Core tasks (read messages, send a voice note, open the assistant) can be completed with audio prompts.
NFR-11	Reliability	Message and report delivery is at least once, with duplicate suppression.
NFR-12	Models	AI model downloads are optional, resumable and removable by the user.

11. Technology stack

Area	Choice	Role
Messaging protocol	Matrix	End-to-end encrypted rooms and events
Homeserver	Synapse with PostgreSQL	Messaging backbone, invitation-token registration
Mobile app	React Native with TypeScript, Android	User interface, offline queue, capture
Native Matrix and crypto module	matrix-rust-sdk bridged to React Native	End-to-end encryption on the device
Local storage	SQLCipher, Android Keystore	One encrypted database per profile
Media provenance	ProofMode and C2PA libraries	Signed capture and proof bundles
Device attestation	Google Play Integrity	Strengthens the captured tier
Duplicate detection	Perceptual hashing in the report service	Flags recycled and re-posted media
Report service	TypeScript or Rust Matrix bot or application service	Report validation, hashing, status handling
Push	Content-free push, optional UnifiedPush	Wake the app without revealing content
Tiv AI serving	Self-hosted inference service	ASR, translation, TTS, assistant
Tiv AI training	Fine-tuning of open speech and translation models	Model development
On-device inference	Compact quantized models	Short-clip ASR, later offline translation
Infrastructure	Docker Compose for development, Terraform later	Reproducible environments
CI	GitHub Actions with a local Synapse for integration tests	Automated build and test

12. Repository and environments

```
/apps/mobile               React Native Android app
/packages/native-matrix    Rust bridge to matrix-rust-sdk
/packages/shared           Shared types: report schema, event types
/services/report-service   Report validation, hashing, duplicate index
/services/tiv-ai           Inference service for ASR, MT, TTS, assistant
/ml                        Data pipelines, training and evaluation code
/infra                     Docker Compose: Synapse, PostgreSQL, push gateway
/docs                      Specifications, threat model, decision records
CLAUDE.md                  Project rules for Claude Code
```

Environments

Environment	Purpose	Data
Development	Local Docker stack for each developer	Synthetic only
Staging	Integration and security testing	Synthetic only
Pilot	Closed trial with invited users	Real, minimal retention

Engineering conventions (CLAUDE.md)

- Never write custom cryptography. Never log message content, keys or identifying data.
- No analytics or third-party SDKs that send data off the device.
- Pin dependency versions and add a dependency only after review.
- Run the full test suite, including profile-separation and wipe tests, before a task is considered done.
- Use the current documentation of Matrix, matrix-rust-sdk, ProofMode and React Native as the source of truth for APIs.
- Record each architectural decision in /docs/decisions.

13. Test plan and acceptance criteria

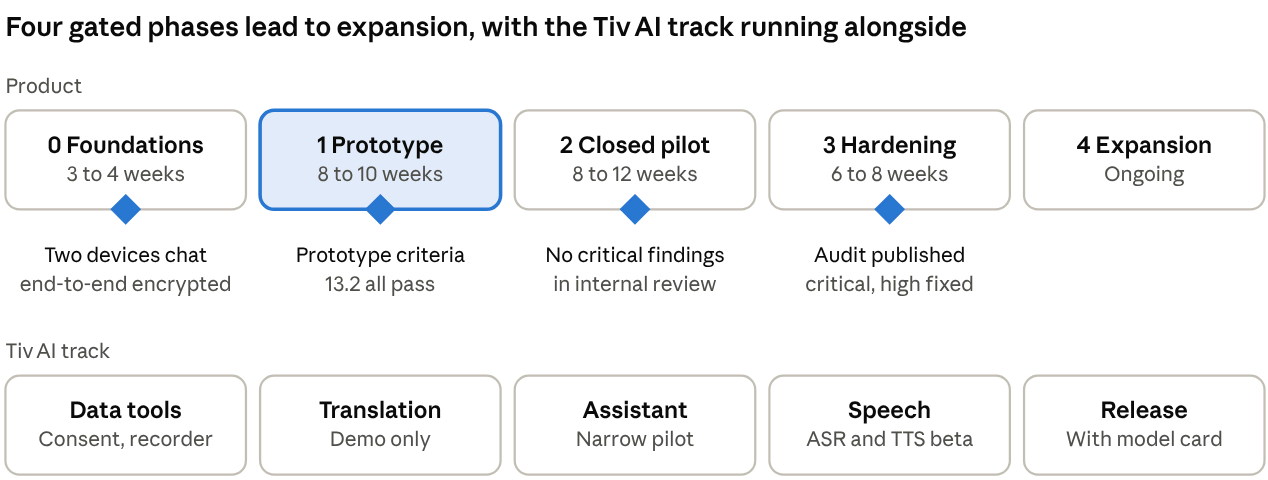
13.1 Test plan

Area	Tests
Messaging	Unit and integration tests against a local Synapse for registration, rooms, messages, voice notes and files
Encryption	Two devices exchange messages, verify each other, rotate keys after a member is removed, and recover after reinstall
Offline	Airplane-mode composing, upload interruption mid-transfer, reconnect and duplicate suppression
Profile separation	Vault data is not readable, listed or loaded before unlock. The duress PIN never exposes vault content in logs, caches, thumbnails or notifications
Wipe	Vault data, keys, caches and queued items are gone after quick wipe
Provenance	Captured media verifies. Edited, re-encoded and imported media fails the captured tier. Known duplicates are flagged
Roles	Each role can do only what the table in section 2 allows
Tiv AI	Held-out evaluation sets for ASR, MT, TTS and the assistant, reviewed by native speakers, plus a red-team set for unsafe advice
Performance	Startup, memory, battery and bandwidth measured on the reference low-cost device
Security	Internal checklist review at the end of the prototype and an external audit before wider deployment

13.2 Acceptance criteria for the prototype

1. Two test devices exchange end-to-end encrypted messages, voice notes and images, including after one goes offline and returns.
2. Registration works only with a valid single-use token and collects no phone number or email.
3. The vault does not appear, load or leave traces before unlock, and the duress PIN shows only social content.
4. A reporter captures a photo, submits a structured report offline, and it arrives on reconnect.
5. A verifier sees the report with proof status, and a resubmitted copy of the same image is flagged as a duplicate.
6. Imported media is labeled unverified and excluded from corroboration.
7. A desk administrator approves an escalation summary that contains none of the excluded fields.
8. Notification payloads contain no sensitive text.
9. Quick wipe removes vault data.
10. The Tiv assistant answers a set of in-domain test prompts through the translation layer, with the evaluation results recorded.
11. All automated tests pass in CI.

14. Delivery roadmap



delivery roadmap · 5 phases, 4 gates, Tiv AI track

Each phase ends at a gate, and the next phase starts when the gate criteria are met. The Tiv AI track runs in parallel and releases publicly in the expansion phase.

15. Glossary

Term	Meaning
Vault	The hidden profile and rooms where reports are submitted and reviewed
Verification desk	The group of verifiers and desk administrators who review reports
Captured tier	Media recorded inside the app and signed at capture
Imported tier	Media brought in from elsewhere, always labeled unverified
Proof bundle	Hash, signature, timestamp and attestation data for a captured item
Perceptual hash	A fingerprint of an image or video frame that stays similar after resizing or re-compression
Corroboration	Independent reports confirming the same event
Escalation summary	A sanitized, desk-approved description of a verified event
Duress PIN	A PIN that opens only the social profile
LGA	Local Government Area
ASR, MT, TTS	Speech to text, machine translation, text to speech
Homeserver	The Matrix server that stores and relays encrypted events